Дифференциатор

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December 15, 2025

1 Результат дифференцирования

$$\left(\frac{\operatorname{arctg}\left(\cos\left(x\right)\right)-x}{\sin\left(x\right)-\operatorname{sh}\left(x\right)}\right)^{\left(x\right)^{5}}\cdot 5\cdot\left(x\right)^{4}\cdot\ln\left(\frac{\operatorname{arctg}\left(\cos\left(x\right)\right)-x}{\sin\left(x\right)-\operatorname{sh}\left(x\right)}\right)+\left(x\right)^{5}\cdot\frac{\frac{\frac{-\sin(x)}{1+(\cos(x))^{2}}-1\cdot\sin(x)-\operatorname{sh}(x)-\operatorname{arctg}(\cos(x))-x\cdot\cos(x)}{(\sin(x)-\operatorname{sh}(x))^{2}}}{\frac{\operatorname{arctg}(\cos(x))-x}{\sin(x)-\operatorname{sh}(x)}}$$

2 Пошаговое дифференцирование

1. $\frac{d}{dx}\left(\cos\left(x\right)\right)$
   $=-\sin\left(x\right)$

2. $\frac{d}{dx}\left(\operatorname{arctg}\left(\cos\left(x\right)\right)\right)$
   $=\frac{-\sin\left(x\right)}{1+\left(\cos\left(x\right)\right)^{2}}$

3. $\frac{d}{dx}\left(\operatorname{arctg}\left(\cos\left(x\right)\right)-x\right)$
   $=\frac{-\sin\left(x\right)}{1+\left(\cos\left(x\right)\right)^{2}}-1$

4. $\frac{d}{dx}\left(\sin\left(x\right)\right)$
   $=\cos\left(x\right)$

5. $\frac{d}{dx}\left(\operatorname{sh}\left(x\right)\right)$
   $=\operatorname{ch}\left(x\right)$

6. $\frac{d}{dx}\left(\sin\left(x\right)-\operatorname{sh}\left(x\right)\right)$
   $=\cos\left(x\right)-\operatorname{ch}\left(x\right)$

7. $\frac{d}{dx}\left(\frac{\operatorname{arctg}\left(\cos\left(x\right)\right)-x}{\sin\left(x\right)-\operatorname{sh}\left(x\right)}\right)$
   $=\frac{\frac{-\sin(x)}{1+(\cos(x))^{2}}-1\cdot\sin\left(x\right)-\operatorname{sh}\left(x\right)-\operatorname{arctg}\left(\cos\left(x\right)\right)-x\cdot\cos\left(x\right)-\operatorname{ch}\left(x\right)}{\left(\sin\left(x\right)-\operatorname{sh}\left(x\right)\right)^{2}}$

8. $\frac{d}{dx}\left(\left(x\right)^{5}\right)$
   $=5\cdot\left(x\right)^{4}$

9. $$\frac{d}{dx}\left(\left(\frac{\operatorname{arctg}\left(\cos\left(x\right)\right)-x}{\sin\left(x\right)-\operatorname{sh}\left(x\right)}\right)^{\left(x\right)^{5}}\right)$$

$$=\left(\frac{\operatorname{arctg}\left(\cos\left(x\right)\right)-x}{\sin\left(x\right)-\operatorname{sh}\left(x\right)}\right)^{\left(x\right)^{5}}\cdot 5\cdot\left(x\right)^{4}\cdot\ln\left(\frac{\operatorname{arctg}\left(\cos\left(x\right)\right)-x}{\sin\left(x\right)-\operatorname{sh}\left(x\right)}\right)+\left(x\right)^{5}\cdot\frac{\frac{\frac{-\sin(x)}{1+(\cos(x))^{2}}-1\cdot\sin(x)-\operatorname{sh}(x)-\operatorname{arctg}(\cos(x}{(\sin(x)-\operatorname{sh}(x))^{2}}}{\frac{\operatorname{arctg}(\cos(x))-x}{\sin(x)-\operatorname{sh}(x)}}$$